

Math 150A Section 8 and 16
Spring 2025
Exam II
March 13, 2025
Time Limit: 1 Hour 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam. However, you may use a single, handwritten, one-sided notesheet and a *basic* calculator.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write

(a) every inflection point of $f(x)$ is also a critical point of $f(x)$

(b) the function $f(x) = x^{1/3}$ has no critical points

(c) if $f(x)$ has three different x -intercepts, then $f'(x)$ must have at least two x -intercepts

(d) the function $-\frac{x^2}{1+x^2}$ is even

(e) if $f''(3) = -2$, then $f(x)$ has a local maximum at $x = 3$

2. (10 points) A grain silo has the shape of an inverted pyramid with a square base. The width of the base is 10 ft and the height of the silo is 30 ft. Grain is poured in at a rate of 10 cubic feet of grain per minute. How fast is the level of the grain in the tank rising at the exact moment when grain level is 3 feet high?

Hint: the volume of a pyramid with a square base b and height h is $V = \frac{1}{3}b^2h$

3. (10 points) Calculate the derivatives of the following functions

(a) $x^3 \cos(x)$

(b) $\frac{x^3+1}{x^2+5}$

(c) $\tan(\sqrt{x})$

4. (10 points) Calculate the derivatives of the following functions

(a) $\sin\left(\frac{2x-3}{x^5+1}\right)$

(b) $(x^8 + 1)^3 \sec(3x + 2)$

5. (10 points) Consider the equation

$$x^9 + 7x^5 + x^5 + 2x^3 + 3x + 4 = 0.$$

- (a) State the Mean Value Theorem

- (b) Use the Intermediate Value Theorem to show that the above equation has at least one solution

- (c) Use the Mean Value Theorem to explain why the equation has no more than one solution

6. (10 points) For each of the following, provide an example or explain why it does not exist.

(a) a function with no critical points

(b) a function with inflection points at -1 and 1

(c) a nonconstant function $f(x)$ whose derivative is always zero

(d) a function with a global maximum at $x = 1$

(e) a critical point of the function $3 - |x - 3|$

7. (10 points) Find the critical points of each of the following functions. Classify them as local maxima, local minima, or neither.

(a) $2x^3 + 9x^2 + 3x + 4$

(b) $9x^{5/3} - 25x^{3/5}$